Deliverable1

October 13, 2026

1 Deliverable 1

1.0.1 Task 1

[84]:
```
import locale
import sys

import matplotlib.dates as mdates
import matplotlib.pyplot as plt
import pandas as pd
import requests
from numpy import mean, median


def load_frost_key(path="../key_frost.py"):
    """Load API client ID from local file."""
    sys.path.append("..")
    with open(path, "r", encoding="utf-8") as file:
        return file.read().strip()


def get_frost_data(client_id, year):
    """Connecting to frost server and get the mean temperatur for 2025."""
    endpoint = "https://frost.met.no/observations/v0.jsonld"
    parameters = {
        "sources": "SN17850",
        "elements": "mean(air_temperature P1D)",
        "referencetime": f"{year}-01-01/{year}-12-31",
    }

    response = requests.get(
        endpoint,
        params=parameters,
        auth=(client_id, "")
    )
    response.raise_for_status()
    return response.json()["data"]
```

```python
def get_the_rigth_format(data):
    """Collect the dates and temperatures from Frost API response and get the␣
 ↪dates in right format."""
    dates = []
    temps = []

    for entry in data:
        date = entry["referenceTime"][:10]  # YYYY-MM-DD
        temp = entry["observations"][0]["value"]
        dates.append(date)
        temps.append(temp)

    df = pd.DataFrame({
        "Dato": pd.to_datetime(dates),
        "Temperatur": temps
    })
    df["Dag"] = df["Dato"].dt.dayofyear
    return df


def summary(df):
    """Creating a summary for min, max, mean and median temperatur."""
    min_temp = df["Temperatur"].min()
    max_temp = df["Temperatur"].max()
    mean_temp = mean(df["Temperatur"])
    median_temp = median(df["Temperatur"])

    min_date = df.loc[df["Temperatur"].idxmin(), "Dato"].date()

    max_date = df.loc[df["Temperatur"].idxmax(), "Dato"].date()
    print("Summary for 2025:\n")
    print(
        f"Min temperature: {min_temp} °C, measured {min_date}\n"
        f"Max temperature: {max_temp} °C, measured {max_date}"
    )
    print(f"Mean: {mean_temp:.1f} °C")
    print(f"Median: {median_temp:.1f} °C\n")



def plot_temperature(df):
    """Plot temperature series with Norwegian month labels."""
    plt.figure(figsize=(12, 5))
    plt.plot(df["Dato"], df["Temperatur"], color="steelblue", linewidth=1.8,␣
 ↪markersize=3)
```

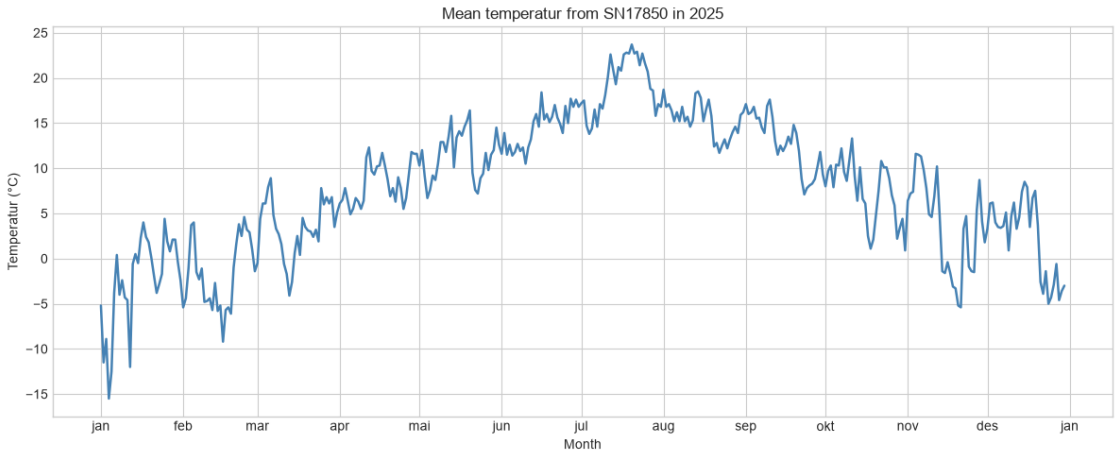
```
    locale.setlocale(locale.LC_TIME, "nb_NO.UTF-8")

    ax = plt.gca()
    ax.xaxis.set_major_locator(mdates.MonthLocator())
    ax.xaxis.set_major_formatter(mdates.DateFormatter("%b"))

    plt.xticks(rotation=0)
    plt.xlabel("Month")
    plt.ylabel("Temperatur (°C)")
    plt.title("Mean temperatur from SN17850 in 2025")
    plt.tight_layout()
    plt.show()


def main():
    """The main program that runs task 1 and collect the right data
    and specifies which data each function shall use."""
    client_id = load_frost_key()
    data = get_frost_data(client_id, 2025)
    df = get_the_rigth_format(data)
    plot_temperature(df)
    summary(df)

if __name__ == "__main__":
    main()
```




```
Summary for 2025:

Min temperature: -15.5 °C, measured 2025-01-04
Max temperature: 23.7 °C, measured 2025-07-20
Mean: 7.9 °C
```

Median: 8.4 °C

1.0.2 Results

We think the graph looks good, and is understandable to read.

1.0.3 Task 2

[85]:
```
def get_frost_data_rain(client_id):
    """Connctin to frost server and get the precipitation for 2025."""
    endpoint = "https://frost.met.no/observations/v0.jsonld"
    parameters = {
        "sources": "SN17850",
        "elements": "sum(precipitation_amount P1D)",
        "referencetime": "2025-01-01/2025-12-31",
    }

    response = requests.get(
        endpoint,
        params=parameters,
        auth=(client_id, "")
    )
    response.raise_for_status()
    return response.json()["data"]

def get_the_rigth_format_rain(data):
    """Collect rain from Frost API response."""
    rain = []

    for entry in data:
        rains = entry["observations"][0]["value"]
        rain.append(rains)

    return rain

def plot_temperature_rain(df, rain):
    """Plot rain series with Norwegian month labels."""

    fig, ax1 = plt.subplots(figsize=(12, 5))

    # plott in the temperature
    ax1.plot(df["Dato"], df["Temperatur"], color="blue", label="Temperature␣
 ↪(°C)")
    ax1.set_xlabel('Month')
    ax1.set_ylabel('Temperature (°C)', color='blue')
    ax1.tick_params(axis='y', labelcolor='blue')
```

```
    # plott in the rain
    ax2 = ax1.twinx()

    ax2.set_ylim(0, 45)
    ax1.set_ylim(-20, 25)

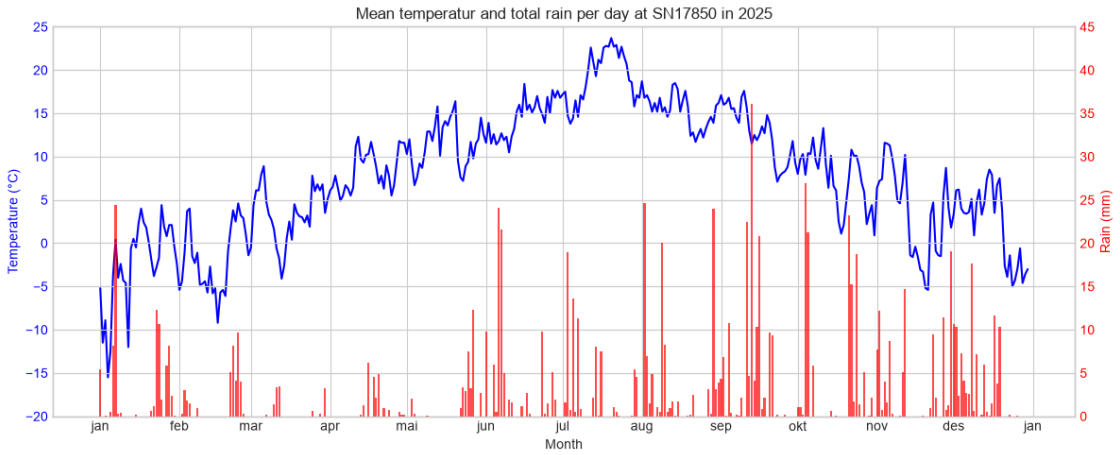
    ax2.bar(df["Dato"], rain, color="red", alpha=0.7, label="Rain (mm)")
    ax2.set_ylabel("Rain (mm)", color="red")
    ax2.tick_params(axis="y", labelcolor="red")


    locale.setlocale(locale.LC_TIME, "nb_NO.UTF-8")
    ax = plt.gca()
    ax.xaxis.set_major_locator(mdates.MonthLocator())
    ax.xaxis.set_major_formatter(mdates.DateFormatter("%b"))


    plt.xticks(rotation=0)
    plt.title("Mean temperatur and total rain per day at SN17850 in 2025")
    fig.tight_layout()
    plt.show()

def main():
    client_id = load_frost_key()
    data = get_frost_data(client_id, 2025)
    rdata = get_frost_data_rain(client_id)
    df = get_the_rigth_format(data)
    rain = get_the_rigth_format_rain(rdata)
    plot_temperature_rain(df, rain)

if __name__ == "__main__":
    main()
```



1.0.4 Results

We chose to use a graph for temperature and bars for rain, we managed to line the grid in the background which makes the plot look good and readable.

1.0.5 Task 3

[86]:
```
def summary2(df):
    """Creating a summary for min, max, mean and median temperatur."""
    min_temp = df["Temperatur"].min()
    max_temp = df["Temperatur"].max()
    mean_temp = mean(df["Temperatur"])
    median_temp = median(df["Temperatur"])

    min_date = df.loc[df["Temperatur"].idxmin(), "Dato"].date()

    max_date = df.loc[df["Temperatur"].idxmax(), "Dato"].date()
    print("Summary for 1925:\n")
    print(
        f"Lowest temperatur: {min_temp} °C, measured {min_date}\n"
        f"Max temperatur: {max_temp} °C, measured {max_date}"
    )
    print(f"Mean: {mean_temp:.1f} °C")
    print(f"Median: {median_temp:.1f} °C\n")


def plot_2_temperature(df_year1, df_year2):
    """Plot temperature series with Norwegian month labels."""
    plt.style.use("seaborn-v0_8-whitegrid")
    plt.figure(figsize=(12, 5))
    plt.plot(df_year1["Dag"], df_year1["Temperatur"], label="1925", linewidth=1.
 ↪8, markersize=3, color="b")
    plt.plot(df_year2["Dag"], df_year2["Temperatur"], label="2025", linewidth=1.
 ↪8, markersize=3, color="r")

    month_names = ("jan", "feb", "mar", "apr", "may", "jun", "jul", "aug",␣
 ↪"sep", "okt", "nov", "des")
    month_starts = (1, 32, 60, 91, 121, 152, 182, 213, 244, 274, 305, 335)
    plt.xticks(month_starts, month_names)
    plt.xlim(1, 365)
    plt.xlabel("Month")
    plt.ylabel("Temperature (°C)")
    plt.title("Compare 1925 mean temperature with 2025 from sources SN17850")
    plt.legend()
    plt.tight_layout()
    plt.show()
```

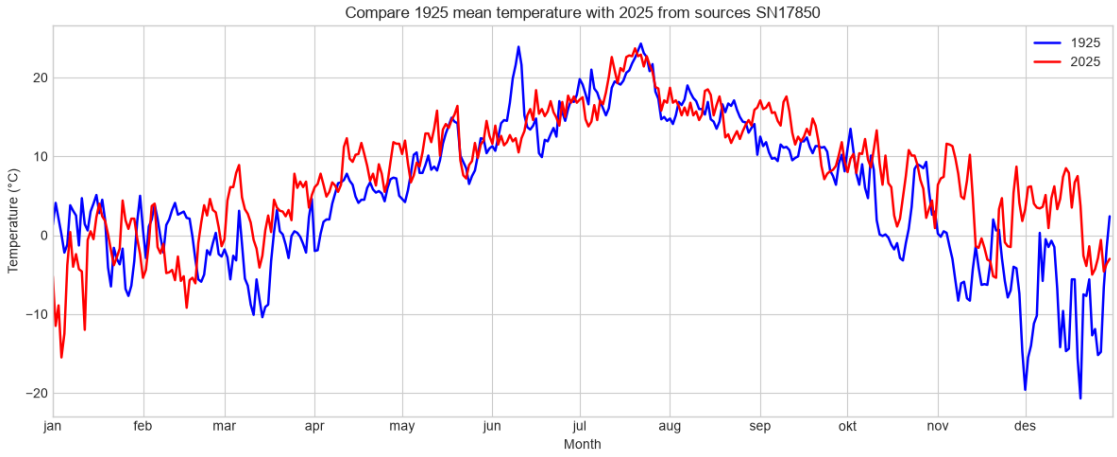
```python
def main():
    """The main program that runs task 3 and collect the right data
    and specifies which data each function shall use."""
    client_id = load_frost_key()

    data_1925 = get_frost_data(client_id, 1925)
    data_2025 = get_frost_data(client_id, 2025)

    df_1925 = get_the_rigth_format(data_1925)
    df_2025 = get_the_rigth_format(data_2025)

    plot_2_temperature(df_1925, df_2025)
    summary2(df_1925)
    summary(df_2025)


if __name__ == "__main__":
    main()
```




```
Summary for 1925:

Lowest temperatur: -20.7 °C, measured 1925-12-20
Max temperatur: 24.3 °C, measured 1925-07-22
Mean: 5.3 °C
Median: 5.3 °C

Summary for 2025:

Min temperature: -15.5 °C, measured 2025-01-04
```

```
Max temperature: 23.7 °C, measured 2025-07-20
Mean: 7.9 °C
Median: 8.4 °C
```

1.0.6 Results

We chose to use two very different colors for the graphs, as you can se the temperature is not that different from 100 years ago, but we dont have data for the trend.

1.0.7 Task 4

[87]:
```python
import yaml


def get_frost_maxdata(client_id, station_id, year):
    """Connect to Frost API and define that we want the max daily temperature␣
 ↪for 2025."""
    endpoint = "https://frost.met.no/observations/v0.jsonld"
    parameters = {
        "sources": station_id,
        "elements": "max(air_temperature P1D)",
        "referencetime": f"{year}-01-01/{year}-12-31",
    }

    response = requests.get(endpoint, params=parameters, auth=(client_id, ""))
    response.raise_for_status()
    return response.json()["data"]


def heatwave (data):
    """Counting how many heatwaves there are in the current period.
    A heatwave is a periode with 5 or more days above 27°C."""

    heatwaves = []
    current_streak = []

    for entry in sorted(data, key=lambda item: item["referenceTime"]):
        day = entry["referenceTime"][:10]
        temp = entry["observations"][0]["value"]

        if temp >= 27:
            current_streak.append(day)
        else:
            if len(current_streak) >= 5:
                heatwaves.append(f"{current_streak[0]} - {current_streak[-1]}")
            current_streak = []
```

```
    if len(current_streak) >= 5:
        heatwaves.append(f"{current_streak[0]} - {current_streak[-1]}")

    return heatwaves


def yml_file(station_id, year, heatwaves):
    """Writes a yml file with the right format"""
    data = {
        "dates": heatwaves,
        "station": station_id,
        "year": year,
        "heatwave_count": len(heatwaves),
    }

    with open("heatwave_summary.yml", "w", encoding="utf-8") as output_file:
        yaml.safe_dump(data, output_file, sort_keys=False)


def main():
    """The main program that runs task 4 and collect the right data
    and specifies which data each function shall use."""
    client_id = load_frost_key()
    station_id = "SN17850"
    year = 2025

    max_data = get_frost_maxdata(client_id, station_id, year)
    heatwaves = heatwave(max_data)
    yml_file(station_id, year, heatwaves)


# We tried to locate the name of the station, so we printed all the elements in␣
 ↪the data, but we didnt find a name
# We found the name on the web page, named "Ås"
#print(max_data[0])



if __name__ == "__main__":
    main()
```

1.0.8 Results

The yaml file we made is in the right format as in the example, but we couldnt manage to get the station_name like in the example. See our code

for an explanation.

1.0.9 Task 5

[88]:
```
def plot_temperature_t5(df):
    """Plot temperature series with Norwegian month labels."""
    df["7_day_av"] = df["Temperatur"].rolling(window=7).mean()
    plt.figure(figsize=(12, 5))
    plt.plot(df["Dato"], df["7_day_av"], color="steelblue", linewidth=1.8,␣
 ↪label="7 day avg")
    plt.plot(df["Dato"], df["Temperatur"], color="red", marker="o",␣
 ↪linewidth=0, markersize =2, label="mean temp that day")

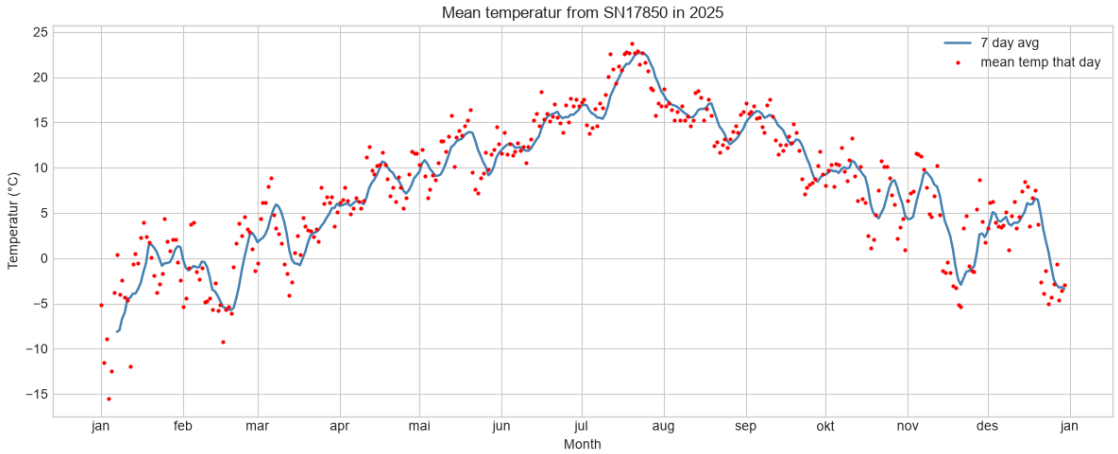
    locale.setlocale(locale.LC_TIME, "nb_NO.UTF-8")

    ax = plt.gca()
    ax.xaxis.set_major_locator(mdates.MonthLocator())
    ax.xaxis.set_major_formatter(mdates.DateFormatter("%b"))

    plt.xticks(rotation=0)
    plt.xlabel("Month")
    plt.ylabel("Temperatur (°C)")
    plt.title("Mean temperatur from SN17850 in 2025")
    plt.legend()
    plt.tight_layout()
    plt.show()


def main():
    """The main program that runs task 5 and collect the right data
    and specifies which data each function shall use."""
    client_id = load_frost_key()
    data = get_frost_data(client_id, 2025)
    df = get_the_rigth_format(data)
    plot_temperature_t5(df)


if __name__ == "__main__":
    main()
```



1.0.10 Results

We managed to make the right graph, it was easy to reuse the data from task 1.

When we follow Pep-8 it was easy to reuse the functions in every task.

1.0.11 sources

Norwegian Meteorological Institute. (2026, 13/10) Frost.met.no.
https://frost.met.no/observations/v0.jsonld

Github repository: https://github.com/GuavaIste/Deliverable-1

1.1 Ai-chatlogs

Task 1 was first written by hand with inspiration from the lectures and
the task was without any functions, we sendt it to ai to get it to pep-8 format.
We used ai to help us with Pandas and format the data.
I wrote a summary and used ai to help me make it better.
We used ai to help make the plots, so they looked good.
Ai knows a lot of very smart and uniqe methods to make coding easier
in a lot of instances. for example task 5 with the .rolling method.

1.2 Video link

https://nmbu.cloud.panopto.eu/Panopto/Pages/Viewer.aspx?id=
10e28da7-e35c-4b21-99ad-b4e200859f9f